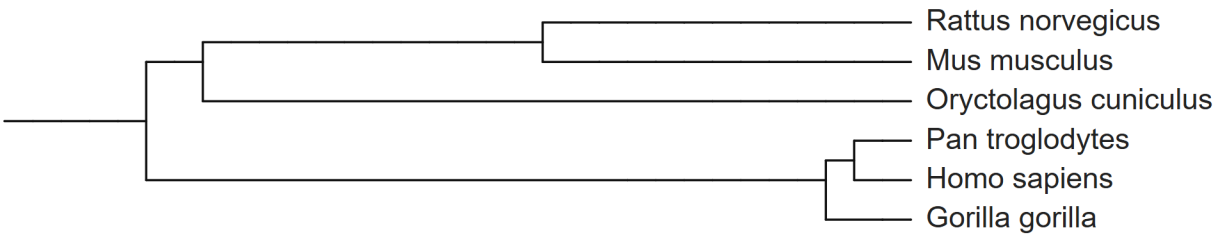
One tree, different drawings

Recognize the same branching relationships across two visual arrangements.

Compare both pages. The source and six selected mammals are identical; only the drawing layout changes. Trace nodes to support your answers.

Name: _____ Date: _____



Rectangular nonmetric cladogram. Upham et al. (2019), S3, Dryad version 4; data CC0.

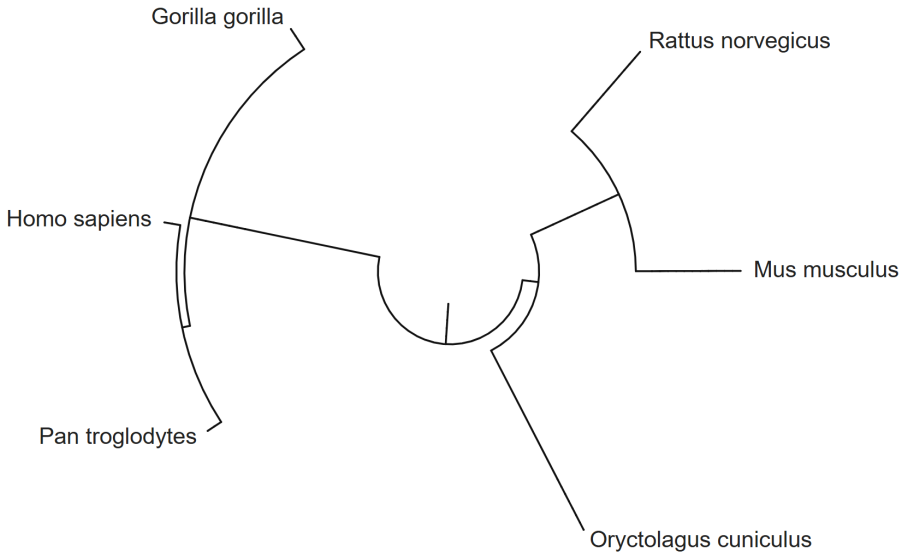
Text description: Both layouts show the same selected source paths. Human and chimpanzee are nested with gorilla; mouse and rat form another represented pair. Taxa omitted from the full source remain omitted in both views.

1. Find the ancestor shared by human and chimpanzee. Locate the larger represented group containing gorilla.

2. Choose another represented pair. Before turning the page, predict whether it will remain a pair in the circular drawing.

Upham, Esselstyn & Jetz (2019). Paper: 10.1371/journal.pbio.3000494. Data: 10.5061/dryad.tb03d03, version 4, S3. Activity: PhyloInk education, 0BSD.

One tree, different drawings - continued



Circular nonmetric cladogram. Same source identities and selection as page 1; data CC0.

Text description: Both layouts show the same selected source paths. Human and chimpanzee are nested with gorilla; mouse and rat form another represented pair. Taxa omitted from the full source remain omitted in both views.

1. Trace the same pairs and nested groups. Has any shared ancestry changed? Explain your evidence.

2. Would exchanging two daughter branches around their shared node change the relationships? Explain without relying on where names appear.

Upham, Esselstyn & Jetz (2019). Paper: 10.1371/journal.pbio.3000494. Data: 10.5061/dryad.tb03d03, version 4, S3. Activity: PhyloInk education, 0BSD.